# Proton Pump Inhibitor Initiation and Withdrawal affects Gut Microbiota and Readmission Risk in Cirrhosis

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OBJECTIVES: **Cirrhosis is associated with gut microbial dysbiosis, high readmissions and proton pump inhibitor (PPI) overuse, which could be inter-linked. Our aim was to determine the effect of PPI use, initiation and withdrawl on gut microbiota and readmissions in cirrhosis.**

METHODS: **Four cohorts were enrolled. Readmissions study: Cirrhotic inpatients were followed throughout the hospitalization and 30/90-days post-discharge. PPI initiation, withdrawal/continuation patterns were analyzed between those with/without readmissions. Cross-sectional microbiota study: Cirrhotic outpatients and controls underwent stool microbiota analysis. Beneficial autochthonous and oral-origin taxa analysis vis-à-vis PPI use was performed. Longitudinal studies: Two cohorts of decompensated cirrhotic outpatients were enrolled. Patients on chronic unindicated PPI use were withdrawn for 14 days. Patients not on PPI were started on omeprazole for 14 days. Microbial analysis for oral-origin taxa was performed pre/post-intervention.**

RESULTS: **Readmissions study: 343 inpatients (151 on admission PPI) were enrolled. 21 were withdrawn and 45 were initiated on PPI resulting in a PPI use increase of 21%. PPIs were associated with higher 30 ($p=0.002$) and 90-day readmissions ($p=0.008$) independent of comorbidities, medications, MELD and age. Cross-sectional microbiota: 137 cirrhotics (59 on PPI) and 45 controls (17 on PPI) were included. PPI users regardless of cirrhosis had higher oral-origin microbiota while cirrhotics on PPI had lower autochthonous taxa compared to the rest. Longitudinal studies: Fifteen decompensated cirrhotics tolerated omeprazole initiation with an increase in oral-origin microbial taxa compared to baseline. PPIs were withdrawn from an additional 15 outpatients, which resulted in a significant reduction of oral-origin taxa compared to baseline.**

CONCLUSIONS: **PPIs modulate readmission risk and microbiota composition in cirrhosis, which responds to withdrawal. The systematic withdrawal and judicious use of PPIs is needed from a clinical and microbiological perspective in decompensated cirrhosis.**

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## INTRODUCTION

Proton pump inhibitors are one of the most highly prescribed medications [1]. In patients with cirrhosis, these medications are used extensively and in many cases, do not have an FDA-approved indication for their continued use [2]. In cirrhotic patients, PPIs are associated with infections, as well as worsening prognosis [2–7]. In compensated cirrhotic patients, PPIs have also been associated with disruptions in the stool microbiota with increase in salivary-origin taxa such as *Streptococcaceae* [8]. However, most infections in cirrhotic patients occur in decompensated cirrhosis and the effect of PPI introduction and withdrawal in these patients is unclear [9]. Other studies have shown that stool

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microbiota in cirrhosis is enriched with these oral origin bacteria but it is not clear whether this is a consequence of cirrhosis or of the widespread acid suppression that is prevalent in this population [8, 10]. This interaction between PPI and infections are relevant since readmissions are a major burden in cirrhosis and changes in PPI use practice could affect this high rate of readmissions [11, 12]. Our aims were to (a) determine the impact of PPI introduction and withdrawal on readmissions in cirrhotic inpatients at 30 and 90 days (b) evaluate the effect of chronic PPI use on the relative abundance of oral-origin taxa in outpatients with cirrhosis and healthy controls in a cross-sectional study and (c) determine the effect of systematic PPI initiation and withdrawal on gut microbial taxa of oral origin in decompensated cirrhotic outpatients in two separate longitudinal studies.

## METHODS

There were four prospective cohorts included in this study. All of these were approved by the Institutional Review Board at Virginia Commonwealth University and McGuire VA Medical Center and all subjects provided written informed consent before study initiation.

### Readmissions study

Inpatients with cirrhosis who were hospitalized for non-elective reasons and were without HIV infection or prior transplant were included. PPI use details (indication, dose and type) at admission, discharge, as well as withdrawal/addition of PPI during the hospitalization were recorded. Patients gave written informed consent, which was obtained from the patients' legally authorized representatives in case they were confused. We only included patients who were discharged alive and without liver transplant to home and had outcomes noted at one and three months post-discharge. The inpatient course, including infections, nosocomial infections, organ failures, acute on chronic liver failure (ACLF), and ICU transfer were recorded in addition to major co-morbid conditions. The major outcomes were readmissions at one and three months from discharge. Variables pertaining to PPI withdrawal/addition during the hospitalization were analyzed. Determinants of readmissions at both time points were analyzed using binary logistic regression with clinical variables (age, gender, MELD score, co-morbid conditions, infections on admission, PPI use change, rifaximin, lactulose, beta-blockers and SBP prophylaxis) that were $p < 0.10$ on univariate analysis. Sample size was calculated based on a prior study with a similar population that showed a 53% 90-day readmission risk [12]. We hypothesized that without PPI this would reduce to 37.5%, which at an alpha of 0.05 and power of 80% would require at least 161 patients per group.

### Cross-sectional microbiota study

In a group of cirrhotic outpatients who could provide informed consent, stool was collected for microbiota composition. Concomitant medications including PPI use, cirrhosis severity and complications were recorded. PPI use was defined as continuous use of PPIs daily for at least 1 month prior to the stool collection, patients using less than this were considered non-users. We also enrolled age and gender matched healthy controls that were on and not on PPI. All controls on PPI were on them for uncomplicated gastro-esophageal reflux disease (GERD). All subjects underwent stool microbiota assessment pertaining to autochthonous taxa [13] (*Lachnospiraceae, Ruminococcaceae*), potentially pathogenic taxa (*Enterobacteriaceae*) and oral-origin taxa (*Streptococcaceae*, *Veillonellaceae*, and *Porphyromonadaceae)* [14, 15]. The cirrhosis dysbiosis ratio (ratio of *Lachnospiraceae* + *Ruminococcaceae* + *Veillonellaceae* + Clostridium Cluster XIV/ *Enterobacteriaceae* + *Bacteroidaceae*) was calculated [15]. Comparisons were performed between patients on and not on PPI. Multi-variable predictors of autochthonous taxa and oral taxa relative abundance were analyzed within cirrhosis.

### Cirrhotic patients studied before and after PPI initiation

A separate cohort of outpatients with decompensated cirrhosis defined by prior variceal bleeding, controlled ascites (on diuretics) and jaundice who were not on PPI were recruited. We excluded patients who were allergic to PPI, those who were unwilling to consent, on treatments for hepatic encephalopathy (rifaximin, lactulose), those on SBP prophylaxis or those on recent (6 weeks) antibiotic or probiotic therapy. All patients who agreed underwent blood and stool collection at baseline and were given 40 mg QD of oral omeprazole for 14 days (Fig. **3a**). A phone call to assess safety, tolerability and adherence was made at day 7 and repeat stool and blood collection was performed at day 14. All subjects were instructed to continue a standard diet. Safety and tolerability, emergence of new cirrhosis complications, MELD score and gut microbiota were analyzed for before/after PPI therapy. Sample size analysis for the cross-sectional study was based on published data in which we found a significant increase in *Streptococcaceae* abundance in compensated cirrhotic patients (0.0 to 9.0%) and in controls (0.0 to 5%) [8]. We assumed that a similar change would be found in controls and cirrhotic patients with/without PPI cross-sectionally, requiring at least 11 subjects per group in either controls with and without PPI using 0.05 level of significance and 80% power.

### Cirrhotic patients studied before and after PPI withdrawal

Another cohort of outpatients with decompensated cirrhosis again defined by defined by prior variceal bleeding, controlled ascites (on diuretics) and jaundice who were on PPI for an indication other than an FDA-approved indication for at least 1 month prior to enrollment were included. FDA approved PPI use indications [16] are (a) Healing and maintenance of erosive esophagitis, (b) Treatment of gastro esophageal reflux disease (GERD), (c) Risk reduction for peptic ulcer with non-steroidal anti-inflammatory drugs (d) Helicobacter pylori eradication, in combination with antibiotics; (e) Pathological hyper-secretory conditions, including Zollinger-Ellison syndrome; (f) Short-term treatment and maintenance of peptic ulcer therapy. We excluded patients who were unwilling to consent or withdraw PPI, on treatments for hepatic encephalopathy (rifaximin, lactulose), those on SBP prophylaxis or those on recent (6 weeks) antibiotic or probiotic therapy. Patients were then asked to withdraw from PPI therapy

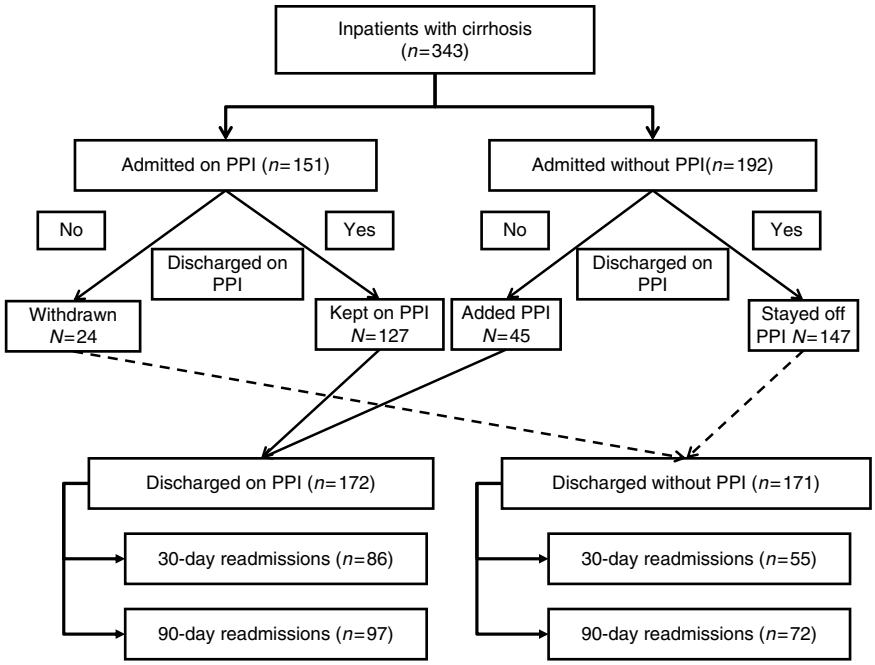



**Fig. 1** Readmissions study, progression of cirrhotic patients through the inpatient study with respect of proton pump inhibitor continuation, withdrawal, and initiation

for 14 days. A phone call to assess safety, tolerability and adherence was made at day 7 and repeat stool and blood collection was performed at day 14. This duration was chosen given prior results in healthy controls and compensated cirrhotic patients [8]. All subjects were instructed to continue a standard diet. Safety and tolerability, emergence of new cirrhosis complications, MELD score and gut microbiota were analyzed for before/after PPI withdrawal.

Sample size analysis for both longitudinal studies was based on published data in which we found a significant increase in salivary *Streptococcaceae* abundance in compensated cirrhotic patients (0.0 to 9.0%) [8]. We assumed that a similar disruption would occur in both situations in which there are PPI compared to the situations without PPI (at baseline for group 1 and end-of-study for group 2) but to a smaller extent due to the baseline dysbiosis in decompensated patients. Using a paired *t*-test at a 0.05 level of significance and assuming a change of 7% and a standard deviation of 8%, a sample size of at least 13 would give us 80% power to detect this effect size.

### Microbiota analysis

*Detailed methods.* Stool samples were analyzed after DNA extraction and multi-tagged sequencing using published techniques (**Supplementary information**) [15, 17]. The unique fraction metric or UNIFRAC measures the differences between microbial communities using the phylogenetic information and is a validated method to determine diversity changes between groups [18, 19]. This was used to compare overall microbial differences between PPI users and non-users in controls and cirrhotic groups. A principal component analysis was performed in the cross-sectional study to determine clustering in controls compared to cirrhotic patients with/without PPI use. In addition, taxa between and within groups were compared using non-parametric approaches (Kruskal–Wallis tests and Wilcoxon matched pairs ranked tests) with a focus on bacterial taxa of interest such as potentially beneficial autochthonous taxa (Lachnospiraceae, Ruminococcaceae) and salivary-origin families (*Porphyromonadaceae, Veillonellaceae* and *Streptococcaceae*).

## RESULTS

### Readmissions study

There were 343 inpatients with cirrhosis recruited prospectively from VCU and Richmond VAMC centers. Of these, 151 patients were on PPI on admission (Fig. 1, Table 1). The reasons for PPI use were GERD ($n=43$), peptic ulcer disease ($n=21$), Barrett's esophagus ($n=18$), concomitant NSAID therapy ($n=7$), after variceal bleeding or after banding ($n=24$), abdominal pain ($n=14$), and rest of the 24 were for unknown reasons. The mean ± SD duration of PPI therapy was 7.45 ± 4.36 years. Most patients were on omeprazole ($n=106$) followed by esomeprazole ($n=23$), pantoprazole ($n=22$), and lansoprazole ($n=3$). Patients on PPI had higher rates of HE, SBP prophylaxis and beta-blocker use compared to those not on PPI but admission MELD score, serum albumin and WBC counts were statistically similar between groups (Table 1). PPI users had a similar rate of most inpatient outcomes but had higher inpatient organ failures compared to non-users. Of these inpatients on PPI, there was withdrawal of PPI in 24 patients and continuation in 127 patients. The reasons for withdrawal were mostly recognition of unnecessary use ($n=16$), change to occasional H2 receptor blockers ($n=5$) and inadvertent stoppage ($n=4$). Of the 192 patients who were

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**Table 1 Inpatients with cirrhosis based on PPI use on admission**

| | Not on PPI N=192 | On PPI N=151 |
|---|---|---|
| Age (years) | 56.5±9.5 | 58.2±9.0 |
| Male Gender | 56 | 44 |
| Etiology (HCV/Alc/HCV+Alc/NASH/others) | 78/21/38/42/13 | 67/10/29/35/10 |
| Admission values | | |
| MELD score | 16.7±7.8 | 17.5±8.3 |
| White blood cell count (/X1000 mm3) | 7.4±4.1 | 7.1±4.1 |
| Serum albumin (g/dl) | 2.7±0.6 | 2.8±0.6 |
| Serum Na (meq/L) | 135.9±5.4 | 135.4±5.7 |
| Admitted for an infection | 40 | 42 |
| Beta-blockers | 66 | 79* |
| Rifaximin | 34 | 56* |
| Lactulose | 72 | 91* |
| SBP prophylaxis | 10 | 21* |
| Co-morbid conditions | | |
| Diabetes | 67 | 48 |
| Cerebrovascular disease | 3 | 2 |
| Cardiovascular disease (Hypertension, Congestive heart failure, coronary artery disease, peripheral vascular disease) | 40 | 36 |
| Chronic obstructive pulmonary disease | 12 | 5 |
| Inpatient course | | |
| GI bleeding during admission | | |
| Intensive care unit transfer | 30 | 30 |
| Second infection | 9 | 12 |
| Mechanical ventilation | 14 | 16 |
| Dialysis | 5 | 14* |
| HE grades 3 or 4 | 20 | 28* |
| Shock | 14 | 20* |
| ACLF | 12 | 17 |
| On Discharge | | |
| MELD score | 18.8±9.3 | 19.5±8.5 |
| Serum albumin | 2.9±0.8 | 2.8±0.7 |
| Beta-Blockers | 66 | 73* |
| Rifaximin | 44 | 63* |
| Lactulose | 99 | 97* |
| SBP prophylaxis | 24 | 27 |
| PPI therapy | 171 | 172 |

Data presented as mean±SD and raw numbers, *$p<0.05$ between groups on Kruskal–Wallis and *t*-tests as appropriate
*ACLF* acute-on-chronic liver failure according to North American Consortium for Study of End-Stage Liver Disease criteria, *PPI* proton pump inhibitor, *HE* hepatic encephalopathy, *SBP* spontaneous bacterial peritonitis, *HCV* hepatitis C virus, *Alc* alcoholic, *NASH* non-alcoholic steatohepatitis

hospitalized without PPI, 45 had PPI added during the hospitalization and 147 remained off PPIs. The 45 patients with PPI addition were due to acid peptic disease ($n=31$), abdominal pain ($n=9$) and post-banding ($n=5$). Most of these 45 patients were discharged on omeprazole ($n=36$) followed by esomeprazole ($n=6$) and lansoprazole ($n=3$). Ultimately 172 patients were discharged on PPI and 172 were discharged without PPI and at the time of discharge, the profile of the patients was largely similar to that of the admission variables.

At one-month post-discharge, we found that 141 patients were readmitted, mostly due to liver disease complications ($n=95$, HE $n=40$, infections $n=28$, renal/metabolic $n=13$ and rest $n=14$). The readmission proportion was significantly higher in those discharged on PPI compared to those who were not ($n=86$, 50% compared to $n=55$, 32%, $p=0.02$). This was also significant on multi-variable analysis where age (OR 0.97, 95% CI 0.94−0.99, $p=0.04$) and PPI use (OR: 1.99, 95% CI 1.28−3.08, $p=0.002$) were predictors of readmission. Other criteria, including admission and discharge MELD score, co-morbid conditions, serum albumin, specific HE, and SBP-related medications and alcoholic etiology of cirrhosis were not additionally contributory to this prediction.

At three months post-discharge, 169 patients (49%) were readmitted. Most of these were due to liver-related reasons ($n=113$, HE $n=45$, infections $n=34$, renal/metabolic reasons $n=16$, upper GI bleeding $n=18$) followed by others (cardiovascular $n=24$, respiratory $n=19$, orthopedic $n=8$, and others $n=5$). Of these 169, a significantly higher proportion of patients on PPI were readmitted at 90 days ($n=97$, 54%) compared to the patients not on PPI on discharge ($n=72$, 41%, $p=0.008$). Patients who were readmitted were also had a similar MELD score on discharge (readmitted 19.9±7.4 vs. not 18.9±10.2, $p=0.27$), rate of discharge beta-blocker use (44% vs. 39%, $p=0.34$), rifaximin use (29% vs. 33%, $p=0.37$), lactulose use (59% vs. 56%, $p=0.63$) and SBP prophylaxis (16% vs. 14%, $p=0.68$). There was a trend towards a lower age of patients who were readmitted, 56.3±8.9 vs. others 58.2±9.7 ($p=0.06$). On univariate analysis, variables studied were age, gender, co-morbid conditions, MELD on discharge, SBP prophylaxis, rifaximin, lactulose, beta-blockers, PPI use and alcoholic etiology of cirrhosis. The only variables with $p<0.10$ were age and PPI use and in a multi-variable analysis, the significant predictors were PPI use (OR 1.6, 95% CI 1.1−2.5, $p=0.03$) and age (OR 0.96, 95% CI 0.94−0.98, $p=0.03$).

Patterns of PPI use were consistent in that all patients on PPI continued their therapy at one and three months while the patients not on PPI remained off PPI in those same time periods. Additionally, 5 patients who were not on PPI were started on H2 receptor blockers.

### Cross-sectional microbiota study

137 cirrhotic outpatients and 45 healthy controls were included. There were no significant differences between groups with respect to age, gender and race. Of these 59 cirrhotic patients were on PPI for an average of 8.2±2.6 years for GERD ($n=16$), peptic disease ($n=12$), NSAID therapy ($n=5$) and unknown reasons ($n=26$).

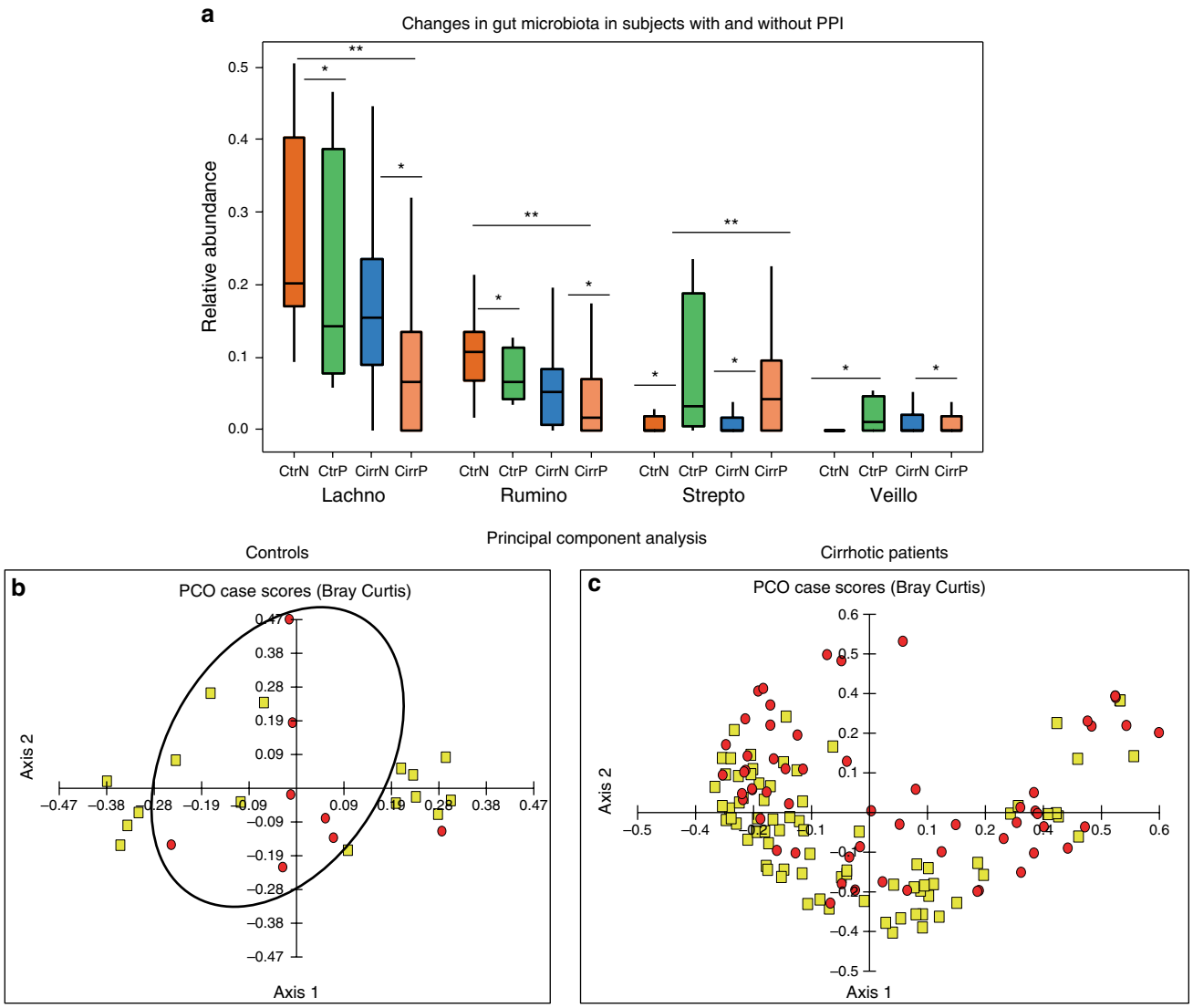



**Fig. 2** Cross-sectional microbiota study, **a** Relative changes in gut microbiota composition of key taxa in outpatient subjects with and without proton pump inhibitor use. Data are presented as median and 95% CI and comparisons are performed using Kruskal–Wallis tests, $*p<0.05$, $**p<0.01$, CtrN: controls not on proton pump inhibitors, CtrP: controls on proton pump inhibitors, CirrN: cirrhotic patients not on proton pump inhibitors, CirrP: cirrhotic patients on proton pump inhibitors, *Lachno: Lachnospiraceae, Rumino: Ruminococcaceae, Strepto: Streptococceae, Veillo: Veillonellaceae*. **b** Principal component analysis of stool microbiota from healthy controls who are on PPI (red) compared to those not on PPI (yellow) showed clustering of the PPI users. **c** Principal component analysis of stool microbiota from cirrhotic patients who are on PPI (red) compared to those not on PPI (yellow) did not show any specific clustering

The leading PPIs used were omeprazole ($n=45$) followed by esomeprazole ($n=12$) and lansoprazole ($n=2$). The 17 healthy controls were on PPI due to GERD for $6.9\pm4.5$ years and were mostly on omeprazole ($n=11$) followed by esomeprazole ($n=6$). Cirrhotic patients on PPI had a higher MELD score and proportion with HE than those who were not (Table 2).

Microbiota analysis showed a significant difference between groups with respect to UNIFRAC ($p<0.001$). There was a significantly lower relative abundance of autochthonous taxa and higher *Streptococcaceae* in PPI users compared to the rest in cirrhotic patients and in healthy controls (Fig. 2a). Interestingly, there was no significant difference in relative abundance of *Streptococcaceae* between controls and cirrhotic patients without PPI. Similarly, the *Streptococcaceae* relative abundance was statistically similar between PPI users regardless of their control or cirrhosis status. On principal component analysis, there was a clustering in controls (Fig. 2a) who were using PPI but not in cirrhotic patients (Fig. 2c). The cirrhosis dysbiosis ratio was significantly lower (i.e., there was more dysbiosis), in PPI users compared to non-users ($0.42\pm0.32$ vs. $0.59\pm0.26$, $p=0.005$). However, the autochthonous family, *Lachnospiraceae* was significantly lower in cirrhotic patients compared to controls regardless of PPI use status. Since within the cirrhosis group, PPI use was higher in those with relatively more advanced liver disease, multi-variable regression

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**Table 2 Outpatients with and without cirrhosis based on PPI use**

| | Controls ($n=45$) | | Cirrhotic patients ($n=137$) | |
|---|---|---|---|---|
| | Not on PPI ($n=28$) | On PPI ($n=17$) | Not on PPI ($n=78$) | On PPI ($n=59$) |
| Age | 54.9±8.9 | 58.8±6.6 | 56.6±6.7 | 56.9±7.1 |
| Male Gender | 71% | 71% | 74% | 75% |
| Race (Caucasian/AA/Hispanic/Others) | 12/10/6/0 | 9/5/4/0 | 54/22/2/0 | 48/10/1/0 |
| Diabetes % | - | - | 37% | 27% |
| Ascites | - | - | 45% | 53% |
| MELD score | - | - | 12.1±6.6 | 15.4±7.2* |
| With varices | | | 23 | 33* |
| Prior HE | - | - | 31% | 54%* |
| Lactulose | - | - | 29 | 21 |
| Additional rifaximin | - | - | 12 | 17* |
| SBP prophylaxis | - | - | 2 | 3 |
| Beta-Blockers | - | - | 22 | 33* |

Data presented as mean±SD and raw numbers unless otherwise indicated, *$p<0.05$ between groups on Kruskal–Wallis and *t*-tests as appropriate
*PPI* proton pump inhibitor, *HE* hepatic encephalopathy, *SBP* spontaneous bacterial peritonitis

models were created for each family including MELD score, HE status, age, gender, alcoholic etiology, PPI use, SBP prophylaxis use, rifaximin use, lactulose use, and beta-blocker use.

Within the cirrhosis cohort, the *Lachnospiraceae* relative abundance was predicted by MELD score ($p<0.0001$) and PPI use ($p=0.02$) independently, while *Ruminococcaceae* was predicted only by MELD score ($p<0.001$). The only factor that significantly predicted *Streptococcaceae* was PPI use ($p=0.03$). The cirrhosis dysbiosis ratio was predicted with the MELD score ($p=0.001$), PPI use ($p=0.02$) and diabetes ($p=0.05$).

Both cirrhotic patients and controls were studied with the variables included being age, control vs. cirrhosis status, gender and PPI use. As in within the cirrhosis group, the only predictor for *Streptococcaceae* relative abundance was PPI use ($p=0.02$) independent of the cirrhosis status. Similarly, *Ruminococcacae* relative abundance was predicted by cirrhosis vs. control status ($p=0.001$). For the *Lachnospiraceae* relative abundance, control vs. cirrhosis status ($p<0.0001$), PPI use ($p<0.0001$) and age ($p=0.005$).

### Cirrhotic patients studied before and after PPI initiation

We enrolled 15 cirrhotic outpatients not on PPI were prescribed omeprazole 40 mg to be taken daily before breakfast for 14 days. All patients had decompensation due to ascites which was well-controlled on diuretics. None of the subjects had current or prior SBP. These patients were then followed for safety, tolerability, as well as microbiota analysis. All patients demonstrated 100% adherence on the medications and did not demonstrate any safety or tolerability concerns. There was no change in the MELD score nor any new complication development during the trial (Table 3). There was a significant increase in the median relative abundance of *Porphyromonadaceae* and *Streptococcaceae* after PPI initiation (Fig. 3b). No changes in *Veillonellaceae* were seen nor changes in other families.

### Cirrhotic patients studied before and after PPI withdrawal

We enrolled 15 patients with decompensated cirrhosis on chronic PPI (all on omeprazole, 12 on 40 mg QD and 3 on 20 mg QD). The decompensation was due to ascites in all patients, which was controlled using diuretics. The patients were on PPI for unapproved indications (abdominal pain in 6, occasional GERD in 5, prior non-peptic bleeding in 2 and unknown indications in 3 patients). No changes in symptoms, including heartburn occurred in between the time when they were off the PPIs. Again, no significant change in MELD score or new complications was seen during this period (Table 3). On microbiota analysis, there was a significant decrease in *Porphyromonadaceae, Streptococcaceae*, and *Veillonellaceae* relative abundance after PPI withdrawal compared to pre-PPI therapy (Fig. 3c).

## DISCUSSION

The results of the current study demonstrate that PPI use is often indiscriminate and initiation and withdrawal of PPIs in decompensated cirrhosis can impact clinical outcomes such as readmissions and change the gut microbial composition. PPI use in cirrhosis has been associated with several negative outcomes in population-based and prospective inpatient/outpatient studies [11]. Few of these studies, however, have focused on readmissions, which have emerged as a major burden on the healthcare system in the US [20]. In a smaller prior study, PPI use was independently associated with newer infection-related readmissions in cirrhotic patients who were included with infections [11]. In a larger study

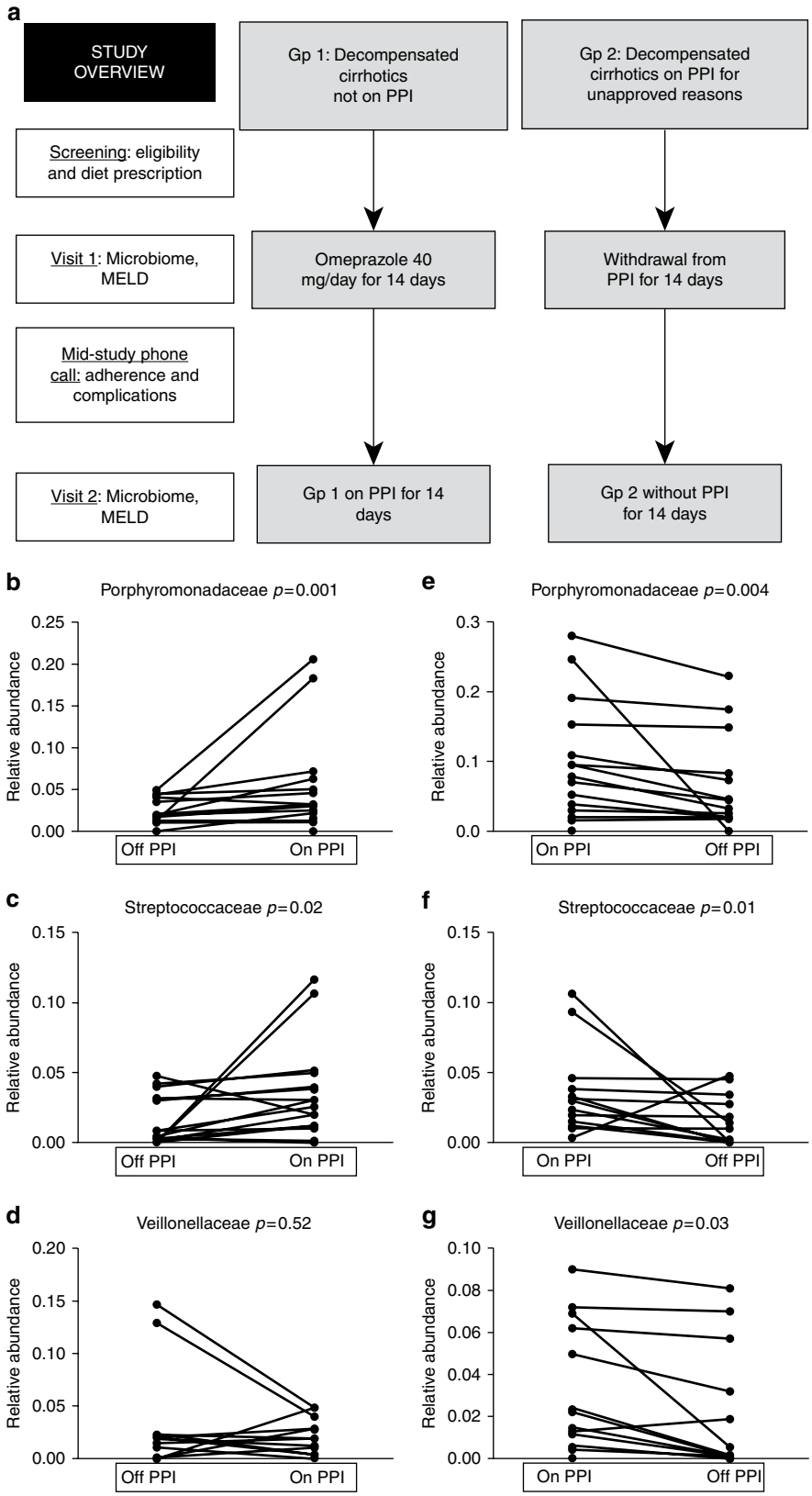



**Fig. 3** Longitudinal studies **a** Schematic of the study design. **b** Before and after omeprazole initiation: Relative abundance of key oral-origin microbiota pre-therapy values compared to post-initiation after 14 days tracked within individuals are presented. Analyses performed using Wilcoxon matched pairs test with p-values displayed in sub-headings. **c** Before and after omeprazole withdrawal: Relative abundance of key oral-origin microbiota while on proton pump inhibitors compared to post-withdrawal for 14 days tracked within individuals are presented. Analyses performed using Wilcoxon matched pairs test with p-values displayed in sub-headings

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**Table 3** Longitudinal studies for PPI initiation and withdrawal in decompensated cirrhosis

| | Initiation study ($n=15$) | | Withdrawal study ($n=15$) | |
|---|---|---|---|---|
| | Baseline not on PPI | 14 days after PPI | Baseline on PPI | 14 days after withdrawal |
| Age | 58±7 | | 59±6 | |
| Male Gender | 14/1 | | 13/2 | |
| Race (Caucasian/AA/Hispanic/Others) | 9/6/0/0 | | 8/5/2/0 | |
| Etiology (HCV, Alc, HCV+Alc, NASH, others) | 6/2/1/4/2 | | 7/2/3/3/0 | |
| Diabetes | 5 | | 6 | |
| MELD score | 13.2±4.5 | 13.5±5.1 | 13.2±5.3 | 12.8±4.9 |
| Ascites | 15 | | 15 | |
| 24 h Caloric intake | 2241±145 | 2308±162 | 2401±236 | 2387±196 |
| Adherence on regimen | – | 100% | – | 100% |

The duration of the trials was 14 days each. None of the patients were on hepatic encephalopathy treatment or SBP prophylaxis throughout the trial. No new or worsening cirrhosis complications or changes in MELD score or dietary intake were noted in both trials
*PPI* proton pump inhibitor, *HCV* hepatitis C virus, *Alc* alcoholic, *NASH* non-alcoholic steatohepatitis

with infected and uninfected patients, PPI use on discharge had a trend towards predicting 90-day readmissions [12]. However, the current study tracks the PPI use including withdrawal as an inpatient, as well as addition of these medications in a granular manner to provide a more detailed insight into their impact on readmissions. While 24 patients of those admitted on PPI were withdrawn from therapy, 45 patients were initiated on PPI, leading to a net increase in PPI therapy on discharge by 15% compared to when these patients were admitted. More importantly, despite having similar MELD scores and concomitant medication use at the time of readmission, PPI-users had a greater readmission rate than non-users. This was independent of usual risk factors for readmission such as MELD score, concomitant medications and alcoholic etiology of cirrhosis, while age was protective. As expected, once discharged on PPI, all patients stayed on PPI therapy or remained off it in those who were discharged without it. The strength of this evaluation is the close monitoring of patients, their dates of initiation and withdrawal of PPI, as well as their readmissions, which were captured completely.

While there are several potential reasons for these negative outcomes in cirrhosis and PPI, the changes in gut microbiota have been hypothesized as a major potential mechanism [21–24]. Prior studies in healthy subjects, compensated cirrhosis, and alcohol-induced rodent models point towards this mechanism but the impact of withdrawal/initiation in advanced or decompensated cirrhosis is unclear [8, 25]. The cross-sectional microbiota study in outpatients with and without cirrhosis demonstrated that potentially beneficial autochthonous taxa such as *Lachnospiraceae* were lower in those with PPI and salivary families such as *Streptococcaceae* were higher in PPI users. Interestingly, *Streptococcaceae* relative abundance was higher in PPI users regardless of whether they had cirrhosis or not, and was only significantly predicted by PPI use on multi-variable analysis. This confirms studies in healthy controls done by our group and others and extends it onto decompensated cirrhosis [8, 21, 23, 24]. In addition, PPI use was associated with *Lachnospiraceae* relative abundance independent of the cirrhosis status of the subject. It is important to note that *Lachnospiraceae* and *Streptococcaceae* but not *Ruminococcaceae*, another autochthonous family, were associated with PPI use. This likely denotes that the PPI effect is not simply a function of advancing liver disease, in which both *Lachnospiraceae* and *Ruminococcaceae* would be affected [15, 26]. When overall microbial change was studied using UNIFRAC, it was significantly different between groups, although clustering on principal component analysis was only seen in control PPI users. This is likely due to the higher impact of PPI in controls who were chosen to be free of other diseases and medications. In contrast, cirrhotic patients had other comorbid conditions, and were likely to be on other medications that could dilute the relative impact of PPI. However, despite this lack of clustering, the cirrhotic PPI users' overall microbiota were different from cirrhotic non-users on UNIFRAC and in the bacterial taxa of interest. Ultimately, the association of the PPI use with a microbial relative abundance pattern that denotes advancing disease (lower cirrhosis dysbiosis ratio), independent of MELD score could be a potential mechanism behind the negative outcomes.

However, within a cross-sectional study, especially in which, like prior studies, the PPI use rate is higher in more advanced patients, the effect of cirrhosis compared to PPI is difficult to delineate [2–4, 27]. Therefore, we performed the two longitudinal studies evaluating PPI introduction and withdrawal over 14 days in decompensated cirrhosis. A similar study of PPI introduction in compensated cirrhosis and healthy controls demonstrated a significant disruption in gut microbiota composition [8]. The population selected was free of HE and antibiotic use but was decompensated due to ascites. Patients with pre-existing HE were excluded due to the impact of HE-related medications on the microbiota that could potentially obscure the impact of PPI therapy [15, 28]. The results demonstrate that oral origin microbiota significantly increased

after PPI initiation, and significantly reduced after the PPI therapy was withdrawn. This is encouraging because it demonstrates that a pattern like compensated cirrhotic patients and controls is followed in decompensated patients and changes related to long-term PPI use can be reversed in as short as 14 days after withdrawal.

The relevance of oral-origin bacteria in cirrhosis is controversial [29]. Prior experience demonstrates that cirrhotic patients have altered salivary microbial composition and that the gut microbiota can increase their relative salivary-origin microbiota relative abundance with only 14 days of PPI therapy [8, 30]. Another report cross-sectionally determined again that cirrhotic subjects have higher oral-origin taxa in their stool [10]. But whether this is a phenomenon related to cirrhosis itself and is complicit in its progression, or is an epiphenomenon of impaired bile acid secretion or gastric acid suppression due to PPI use or cirrhosis itself is an important question. Our results demonstrate that this is likely to be affected significantly by PPI use, since withdrawal of PPIs resulted in a dramatic reduction and vice versa, even in decompensated cirrhosis. Given the some *Streptococcal* species can cause SBP; they may not be entirely innocent bystanders in this situation [31]. In addition, *Porphyromonadaceae* has been associated with systemic inflammation, and brain dysfunction associated with HE [32, 33]. The lower relative abundance of *Lachnospiraceae* with a concomitant *Streptococcaceae* increase could potentially indicate that the oral origin taxa could be crowding out the beneficial taxa, which in turn could also aid in cirrhosis progression. Therefore, the increase in oral origin taxa in the gut microbiota by PPI in cirrhosis is indicative of a major disruption and could be a marker for worse prognosis. The mechanism of these changes are unclear but a direct impact of PPI on immune capability, in addition to the relatively increased abundance of these taxa due to acid suppression could be hypothesized to synergize in this situation [34].

The study is limited by a relatively small number of subjects who were withdrawn from PPI as inpatients. However, most of the reasons for withdrawal were related to a greater awareness of the negative impact of unnecessary PPI use and will likely increase with greater education. We also excluded HE patients from the longitudinal studies and studies including them in larger sample sizes are required. We only studied the stool but not the mucosal microbial taxa, which could be changed differentially. We also did not find a significant effect of antibiotics (SBP prophylaxis and rifaximin) on the multi-variable analysis of microbiota, but this could be related to the smaller sample size of patients on SBP prophylaxis and the predominant effects of rifaximin on microbial function rather than composition [28]. We also assumed that all PPIs would have similar effects on outcomes and microbiota, however, most patients were on omeprazole and this agent was used for the initiation study as well.

The findings of this study demonstrate that PPI use patterns change significantly during hospitalizations, which can impact 30 and 90-day readmission risk independent on cirrhosis severity, hepatic encephalopathy and concomitant medications. Chronic PPI use is associated with lower autochthonous and higher oral origin taxa in cirrhotic outpatients, which is independent of cirrhosis severity and concomitant medications. These increases in oral origin taxa can be brought upon by a two-week PPI course and reversed during the same period even in decompensated cirrhosis. The systematic withdrawal and judicious use of PPI should be encouraged from a clinical and microbiological perspective in decompensated cirrhosis. Further studies of PPI withdrawal in larger populations to study this systematically are needed.



### CONFLICT OF INTEREST

**Guarantor of the article:** JSB is the guarantor.

**Specific author contributions:** JSB conceptualized the study, MS and PMG were involved in microbiota analysis, PBH was involved in the critical revisions, CA, MBW, AF, and EAG were involved in research coordination, Rest of the authors were involved in subject recruitment and follow-up.

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**Potential competing interests:** None.

## Study Highlights

### WHAT IS KNOWN

- Readmissions are a major healthcare burden in cirrhosis and current clinical data are often inadequate to predict this.
- Proton pump inhibitor use is epidemic, is usually due to unindicated reasons in advanced cirrhosis, and is often associated with negative outcomes.
- The impact of proton pump withdrawal and initiation in the inpatient setting on readmissions is unclear
- Proton pump inhibitor use can increase in oral-origin microbiota in the stool after short-term treatment in compensated cirrhosis, which could affect these outcomes.
- However, the effect of proton pump inhibitor withdrawal and initiation in decompensated cirrhosis, who experience many of the negative outcomes in cirrhosis, needs to be studied

### WHAT IS NEW HERE

- In inpatients with cirrhosis, almost a fifth of patients are either initiated or withdrawn from proton pump inhibitors, which can predict 30 and 90-day readmissions independent of cirrhosis severity and concomitant medications.
- Healthy controls and outpatient cirrhotic patients on chronic proton pump inhibitor therapy have a higher relative abundance of oral-origin taxa and in cirrhotics, associated with lower potentially beneficial autochthonous taxa.
- In decompensated cirrhotic outpatients not on proton pump inhibitors, omeprazole initiation and use over two weeks increases oral-origin taxa, while in another group of decompensated cirrhotic outpatients on chronic proton pump inhibitors, the baseline high oral-origin microbiota relative abundance is reversed after omeprazole is withdrawn for two weeks.

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